

HyperGranular-RAG: Structured Evidence Completion for Compact Multi-Hop Retrieval

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Abstract

Multi-hop retrieval-augmented generation must assemble complementary evidence within a fixed context budget, yet relevance-only rankings may omit bridge evidence and unconstrained expansion may displace useful items. We introduce HyperGranular-RAG, an inference-time evidence-completion framework that organizes candidate sentences into adaptive granular balls, connects query-relevant balls with facet hyperedges, and inserts a small number of candidates after a protected dense prefix. Across three closed-candidate HotpotQA and MuSiQue evaluations with Qwen2.5-1.5B-Instruct, HyperGranular-RAG improves answer F1 over a compact MiniLM retriever by +0.01478 [0.00020, 0.02988], +0.01140 [0.00450, 0.01835], and +0.01357 [0.00491, 0.02233]. A matched ablation further shows a +0.01336 [0.00341, 0.02343] gain for facet-conditioned selection over centroid-only expansion. With BGE large-en-v1.5, the original system underperforms the strong dense baseline, while cross-space and BGE-native extensions yield no statistically resolved improvement. Holding the inserted set fixed, protected placement improves over unprotected placement. These findings show that structured evidence completion can recover evidence gaps for a compact retrieval backbone, while its marginal value contracts with a stronger retriever. Paired bootstrap intervals, pre-specified evaluation settings, and independent artifact verification support these scoped conclusions.

1 Introduction

Retrieval-augmented generation (RAG) gives language models access to non-parametric evidence at inference time (Lewis et al., 2020; Guu et al., 2020). Dense passage retrieval provides an effective learned relevance ordering (Karpukhin et al., 2020), but multi-hop questions introduce

a joint-coverage requirement: the selected context must contain multiple pieces of evidence, not merely high-scoring passages considered independently. HotpotQA and MuSiQue make this requirement explicit through multi-hop questions and supporting-evidence annotations (Yang et al., 2018; Trivedi et al., 2022).

Under a fixed Top- k budget, recovering a missing bridge necessarily competes with evidence already selected by the retriever. If individually similar units occupy much of the ranking, a complementary bridge may remain outside the generator context. Expansion can recover that bridge, but every insertion can displace an item that the base retriever ranked highly. Retrieved evidence is also not uniformly useful to a generator: position, redundancy, prompt construction, and the generator itself affect whether a retrieved item changes the answer (Liu et al., 2024; Ram et al., 2023). Evidence coverage, ranking placement, and answer utility must therefore be measured separately.

HyperGranular-RAG is a structured evidence-completion layer for dense retrieval under this fixed context budget. It organizes sentence units into adaptive granular balls, identifies cross-ball relations through query-aware facet hyperedges, and inserts at most four eligible units after a protected Dense Top-10 prefix while preserving an effective Top-20. After development-time configuration, retrieval on confirmation data proceeds without access to answers, supporting-fact labels, generator outputs, or downstream scores. Development labels are used only for designated configuration selection and remain separate from confirmation evaluation. Granular balls act as adaptive local clusters in embedding space, while hyperedges represent higher-order relations involving more than pairwise adjacency (Xia et al., 2019; Feng et al., 2019).

We evaluate whether structured evidence completion improves compact dense retrieval and

whether the same mechanism provides additional value with a stronger retriever. The study covers pre-specified HotpotQA and MuSiQue closed-candidate settings, matched component contrasts, one additional generator configuration, a BGE large-en-v1.5 baseline, a cross-space BGE sidecar, and a BGE-native reconstruction. Development choices remain separate from confirmation, and independent verifiers reconstruct the reported comparisons.

Our study yields three main findings. First, static HyperGranular-RAG repeatedly improves answer F1 over the compact MiniLM Dense backbone in the three Qwen evaluations. Second, facet-conditioned selection improves over the specific centroid-only NoFacet ablation within that system. Third, when the cross-space sidecar holds the inserted candidate set constant, protected placement improves over direct unprotected placement. The gains are backbone-dependent: the original system falls below BGE, and neither tested BGE extension yields a statistically resolved answer-quality improvement. Our conclusions are restricted to the evaluated closed-candidate settings; open-domain retrieval, broader generator transfer, and the independent necessity of granular balls remain open questions.

We make three contributions:

- We introduce a structured evidence-completion layer that combines adaptive granular-ball organization, query-aware facet hyperedges, and protected budget-constrained insertion under a fixed Top- k budget.
- Through controlled HotpotQA and MuSiQue evaluations, we establish repeatable answer-F1 gains over a compact MiniLM retriever and matched evidence for facet-conditioned selection over centroid-only expansion.
- Through component, placement, generator-transfer, and strong-retriever studies, we characterize when structural completion provides marginal value and when that value contracts, with independently verifiable artifacts supporting the reported scope.

2 Related Work

Dense and multi-hop retrieval. Dense passage retrieval learns dual-encoder representations for efficient passage ranking (Karpukhin et al., 2020).

Multi-hop variants extend retrieval from independent passages to evidence chains. Multi-Hop Dense Retrieval conditions later retrieval on preceding evidence (Xiong et al., 2021), while BeamDR uses beam search over dense reasoning paths (Zhao et al., 2021). PullNet iteratively expands from retrieved entities over text and knowledge bases (Sun et al., 2019), and IRCOT interleaves retrieval with chain-of-thought reasoning (Trivedi et al., 2023). HyperGranular-RAG is not an iterative question-rewriting method. Within each fixed closed candidate set, it performs a single bounded completion of an existing ranking.

Graph, hypergraph, and hierarchical retrieval.

Structure-aware retrieval represents relations that a flat relevance ordering can miss. RAPTOR retrieves from a tree of recursively summarized text (Sarathi et al., 2024). HippoRAG combines a knowledge graph with personalized propagation (Gutiérrez et al., 2024), and GNN-RAG uses graph-neural retrieval over knowledge graphs (Mavromatis and Karypis, 2025). GraphRAG targets query-focused summarization using graph communities (Edge et al., 2024). HyperGranular-RAG differs in its evidence unit, relation, and budget contract: adaptive balls organize local sentence units; facets define query-conditioned cross-ball hyperedges; and insertion preserves a dense prefix instead of replacing the complete ranking. These graph systems address broader construction and retrieval settings, whereas our study isolates bounded completion within a fixed candidate set; direct comparison is left to future work.

Evidence expansion and generation. Retrieval-enhanced models can expose external evidence, but the generator still consumes a finite context. RECOMP studies selective and compressed augmentation (Xu et al., 2024), while long-context studies show that evidence position affects model use even when the information is present (Liu et al., 2024). Fusion-in-Decoder demonstrates the importance of how multiple passages are consumed (Izacard and Grave, 2021), and in-context RALM shows that retrieved text can condition language models without parameter updates (Ram et al., 2023). We therefore use paired answer F1 as the primary endpoint and treat retrieval coverage as supporting evidence.

Alternative completion controls. A simpler completion rule could combine relevance with di-

versity or uncovered query-term coverage without constructing higher-order relations. The NoFacet arm in this study is narrower: it preserves granular-ball geometry and selects up to two eligible balls by query-to-centroid score. It is therefore a matched test of the pre-specified facet-conditioned selector against centroid-only ball expansion, not a comparison with all relevance–diversity or coverage objectives. We retain this distinction in every component claim.

Evaluation and negative evidence. KILT emphasizes provenance-aware evaluation in knowledge-intensive tasks (Petroni et al., 2021). Reproducibility guidance motivates explicit dependencies, artifacts, and reporting boundaries (Pineau et al., 2021). Statistical uncertainty is not resolved by relabelling an interval crossing zero as “no effect” or equivalence (Amrhein et al., 2019). We therefore report comparisons as supported, negative, statistically unresolved, or without a matched estimand.

3 HyperGranular-RAG

3.1 Problem Formulation

For a query q , let $U_q = \{u_i\}$ be the closed candidate set of sentence units. A fixed encoder maps the query and each title + “. ” + sentence unit to an L2-normalized representation. The base relevance score is cosine similarity,

$$s(q, u_i) = \hat{e}_q^\top \hat{e}_i. \quad (1)$$

Ties are resolved by ascending `unit_id`. The Dense baseline returns the first $K_q = \min(20, |U_q|)$ unique units. HyperGranular-RAG constructs an additional ordered set of eligible units but must return the same effective K_q . No answer, gold supporting unit, or generator output enters this construction.

3.2 Adaptive Granular Balls

Each query’s candidate embeddings begin as one root set. For a current ball B , let

$$c_B = \text{norm} \left(\frac{1}{|B|} \sum_{i \in B} \hat{e}_i \right), \quad d_i = 1 - \hat{e}_i^\top c_B, \quad (2)$$

with radius $r_B = \max_{i \in B} d_i$ and compactness $1 - |B|^{-1} \sum_{i \in B} d_i$. In the compact MiniLM variant, B is split iff its depth is below 6, $|B| \geq 4$, and either $|B| > 3$ or $r_B > 0.78$. The first seed is the

unit farthest from c_B ; the second has minimum cosine similarity to the first. Each unit is assigned to the seed with higher cosine similarity. A split is retained only when both children contain at least two units. Ties preserve the candidate order. We keep these parameters fixed across evaluation settings; they are not learned from confirmation data.

3.3 Query-Aware Facet Hyperedges

Each ball aggregates content terms from titles and sentences after fixed ASCII-alphanumeric and stop-word rules. Its facet terms are the intersection between ball terms and query terms. The two balls with highest query-to-centroid cosine become seeds. A non-seed ball can form an expansion hyperedge only if it introduces a new query term and passes fixed gates on ball size, ball score or seed proximity, units per new term, and shared-term redundancy. We use *facet hyperedge* for this query-conditioned relation between the seed balls and an expansion ball that contributes new query-term coverage.

For a candidate ball B , let n_B be the number of query terms newly covered beyond the seeds, t_B the total number of query terms in B , m the number of query terms, $b_B = \hat{e}_q^\top c_B$, a_B the maximum cosine to a seed centroid, and ρ_B the fraction of B ’s facet terms already covered. The compact-variant score is

$$\begin{aligned} h_B = & 0.45 n_B / \max(m, 1) \\ & + 0.20 t_B / \max(m, 1) \\ & + 0.20 b_B + 0.15(1 - a_B) \\ & - 0.20 \rho_B - 0.05 |B| / 8. \end{aligned} \quad (3)$$

A candidate must introduce at least one new term, contain at most 8 units, satisfy $b_B \geq 0.12$ or $a_B \geq 0.05$, have at most 6 units per new term, obey $\rho_B \leq 0.90$, and have $h_B \geq 0.10$. Candidates are ordered by descending h_B and then `edge_id`; at most two distinct balls are selected. Units inside an accepted ball remain ordered by original MiniLM cosine and `unit_id`; the fixed unit-level facet bonus is zero. Consequently, the hyperedge selects candidate regions rather than learning a new dense ranker.

3.4 Protected Bounded Insertion

The compact variant protects the first ten unique Dense units. HyperGranular-RAG units whose cosine is at least 0.1957079917192459 are considered in the pre-specified order. At most four candidates absent from the protected prefix are inserted

immediately after that prefix. Remaining Dense units are appended in their original order, followed only if necessary by unfilled expansion units, and the result is truncated to K_q .

We set the candidate-score floor to 0.1957079917192459, the 25th percentile selected during earlier development, and keep it fixed in all subsequent evaluations (Table 4). We do not treat it as an independently optimized value.

Protected insertion reduces the risk of displacing highly ranked evidence, but it does not guarantee that an inserted candidate is useful. This distinction motivates the placement and BGE comparisons below.

3.5 Compact, Sidecar, and Native Variants

The *compact variant* uses MiniLM embeddings for the base ranking, ball geometry, facet gates, and unit ordering. The *cross-space sidecar* keeps BGE Top-20 as the primary ranking while using the fixed MiniLM-HyperGranular-RAG structure to propose eligible candidates; no score fusion is introduced. The *BGE-native variant* reconstructs candidate geometry and structural selection in BGE space under one configuration selected during development and then fixed for confirmation. We evaluate the three variants separately because each isolates a different interaction between structural completion and the base retriever.

With embeddings precomputed, the closed-candidate procedure is $O(H|U_q|d + |U_q|\log|U_q| + |\mathcal{B}_q|\log|\mathcal{B}_q|)$; Appendix A states the component bounds and memory cost.

4 Experiments

4.1 Datasets and Evaluation Settings

HotpotQA provides diverse multi-hop questions and supporting facts (Yang et al., 2018); MuSiQue composes single-hop questions to reduce shortcut solutions (Trivedi et al., 2022). We evaluate retrieval within the candidate sets supplied by each benchmark; corpus-scale indexing is outside the scope of this study.

We evaluate the compact variant on 1,000 HotpotQA queries, 3,000 MuSiQue queries, and a separate component set with 1,000 HotpotQA and 1,500 MuSiQue queries. The cross-space sidecar and BGE-native confirmation studies each use a separate, zero-overlap set with 1,000 HotpotQA and 1,500 MuSiQue queries. Query selection is de-

terministic, and we verify dataset and query identities.

4.2 Retrievers and Generators

The compact baseline is the fixed MiniLM dense retriever used throughout the original development program. BGE large-en-v1.5 is the pre-specified strong dense baseline; the broader BGE family is documented by Xiao et al. (2023), while the exact model revision is bound in repository manifests.

The main generator is Qwen2.5-1.5B-Instruct FP16 with a fixed short-answer RAG prompt and a 4,096-token input cap. One additional transfer evaluation uses google/gemma-4-E2B-it-qat-mobile-transformers in its official mobile-QAT format. Because Qwen and Gemma use different numerical formats and runtimes, this comparison characterizes the tested deployment configurations rather than intrinsic architectural quality or cross-device efficiency.

4.3 Experimental Design

We specify all method and decision parameters before evaluating the corresponding confirmation split. The BGE-native development study selected configuration C10 with a development answer-F1 difference of +0.003143. We use this development result only to select C10 and report its confirmation performance separately. The pre-registered confirmation outcome ended the BGE-native configuration search.

4.4 Metrics and Statistical Decisions

The primary endpoint is paired query-level answer-F1 difference. Answer exact match (EM), retrieval complete recall at 20 (CR@20), and evidence recall at 20 (ER@20) are supporting endpoints. Each comparison uses 10,000 paired bootstrap samples and pre-specified decision rules. Dataset-joint comparisons assign equal weight to HotpotQA and MuSiQue. An interval crossing zero is statistically unresolved and is not interpreted as equivalence because no equivalence margin or formal equivalence test was specified. Sample sizes define available resource and interval precision, not a formal power guarantee.

4.5 Label Access and Evaluation Integrity

On confirmation data, candidate construction, retrieval, prompting, and generation proceed without access to answer labels, supporting-fact annotations, or evaluation feedback. Designated devel-

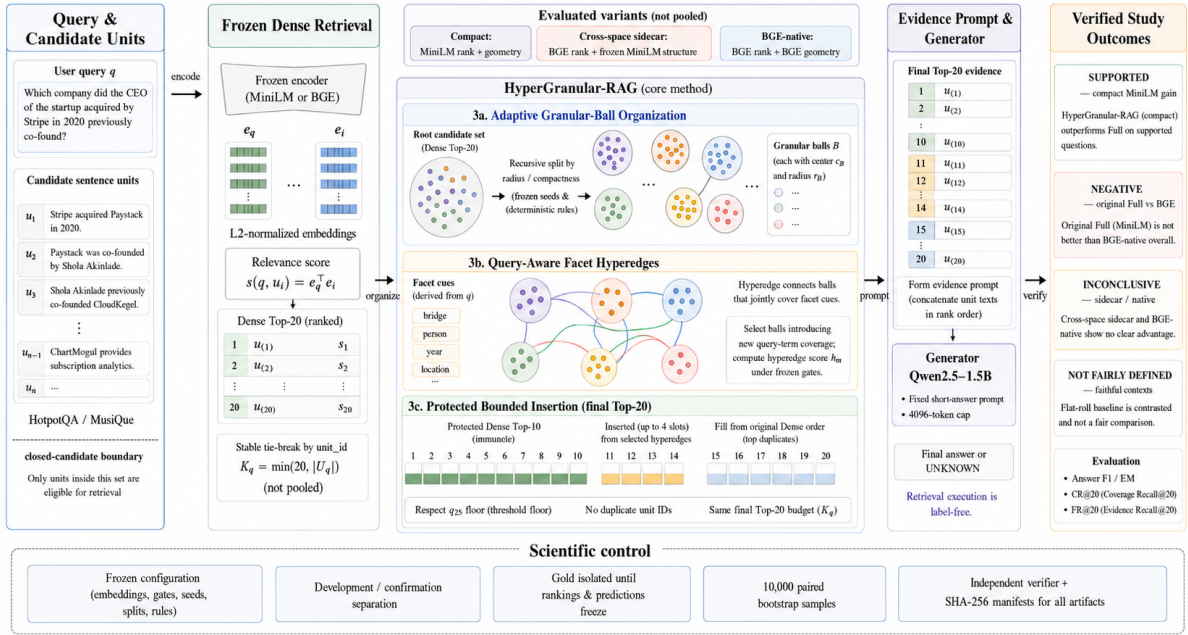


Figure 1: **Overview of HyperGranular-RAG and the evaluated retrieval variants.** Candidate units are ranked by a fixed dense retriever, organized into adaptive granular balls, connected through query-aware facet hyperedges, and inserted under a protected, bounded Top- k policy before answer generation. Compact, cross-space sidecar, and BGE-native variants are evaluated separately rather than pooled. The right column distinguishes supported, negative, statistically unresolved, and unmatched outcomes, while the bottom row summarizes the scientific controls applied to the verified experiments.

opment labels may select a configuration, including BGE-native C10, whereas confirmation labels remain inaccessible until rankings, prompts, and predictions are fixed and are not reused for selection. Thus “label-free” refers to retrieval execution rather than the entire research process. Evaluators verify gold identity before parsing it, and independent verifiers reconstruct ranking order, metrics, bootstrap summaries, decisions, and artifact identities. Main/run identity is checked in full or on a pre-hash deterministic subset.

5 Results

5.1 Repeated Gains over Compact Dense Retrieval

Table 1 and Figure 2 show three Qwen comparisons. On 1,000 HotpotQA queries, FULL minus DENSE answer F1 is +0.01478 [0.00020, 0.02988]. On 3,000 MuSiQue queries, the difference is +0.01140 [0.00450, 0.01835]. The separate joint component evaluation again supports FULL minus DENSE with a dataset-equal-weight difference of +0.01357 [0.00491, 0.02233].

HyperGranular-RAG improves answer F1 in all three compact-retriever evaluations, with effect sizes from +0.01140 to +0.01478. These gains

apply to the MiniLM backbone, Qwen generator, fixed prompt, closed candidates, and Top-20 budget; the strong-retriever experiments below evaluate whether the pattern extends to BGE.

5.2 Component Evidence

Within the fixed MiniLM system, FULL minus No-Facet is +0.01336 [0.00341, 0.02343], supporting an incremental role for facet conditioning relative to the specific centroid-only ball selector. This matched contrast isolates facet conditioning within the granular-ball candidate pool; generic relevance–diversity and coverage selectors remain outside the present comparison. FULL minus No-Protection is +0.00354 [−0.00675, 0.01389], so the independent contribution of protection is statistically unresolved in that setting. We do not report a flat-unit granular-ball ablation because ball seeds, split gates, edge budgets, and radii have no unique matched counterparts at the unit level. This missing matched estimand provides no evidence for or against a granular-ball effect.

In the cross-space sidecar, the inserted set is identical between protected and unprotected conditions. PROTECTED minus Unprotected is +0.01122 [0.00129, 0.02104], which supports the

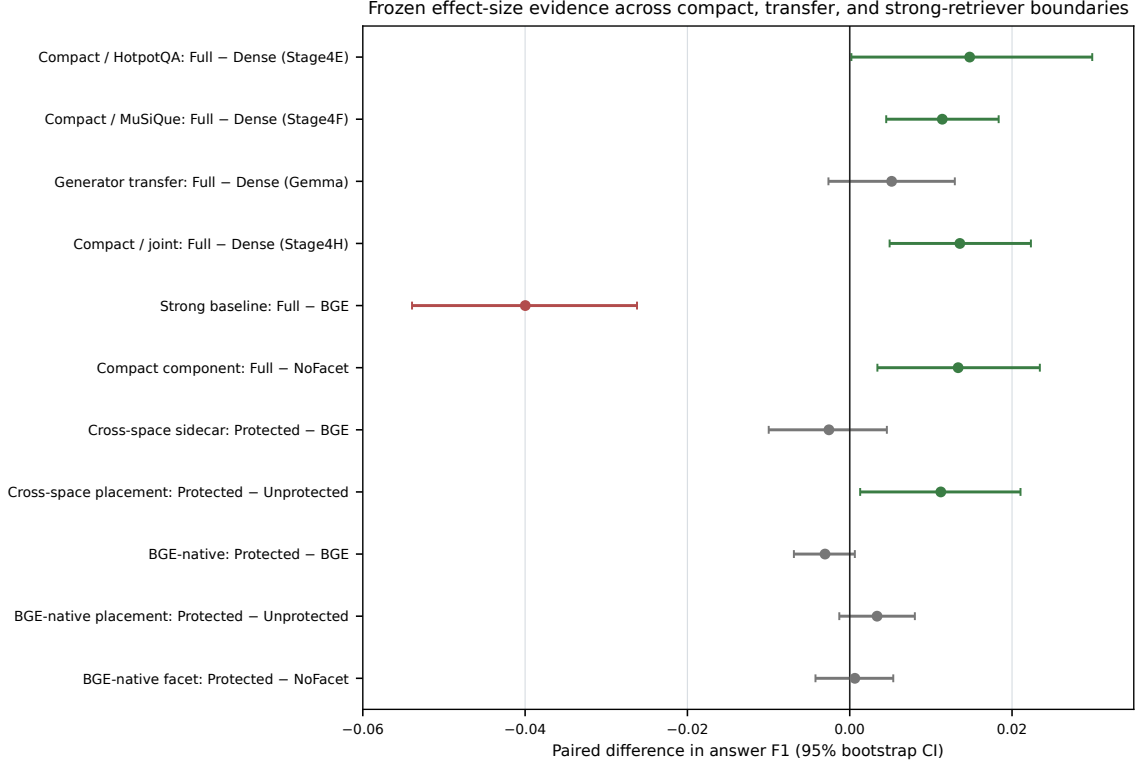


Figure 2: **Paired answer-F1 effect sizes.** Points are paired answer-F1 differences and bars are pre-specified 95% bootstrap confidence intervals. The zero line is explicit. Comparisons whose intervals cross zero are statistically unresolved rather than evidence of equivalence.

Setting	F1 D/F	EM D/F	CR@20 D/F	ER@20 D/F	Avg. ins.	F1 G/H	Δ F1 [95% CI]
HotpotQA	.42150/.43628	.35600/.36600	.73700/.79800	.87860/.90818	1.9300	47/34	+0.01478 [.00020, .02988]
MuSiQue	.13595/.14735	.10033/.11033	.58900/.65000	.80553/.84128	2.4110	125/77	+0.01140 [.00450, .01835]
Joint	.30446/.31803	.25467/.26400	.66617/.72217	.84529/.87657	2.1628	102/73	+0.01357 [.00491, .02233]

Table 1: Absolute compact-dense outcomes and paired effects. D/F denotes DENSE/FULL; G/H counts queries with positive/negative answer-F1 change. Average insertions are structural candidates placed per query. The Joint row is dataset-equal-weighted for absolute metrics; G/H and insertions use all 2,500 queries. All G/H and insertion counts are descriptive reconstructions from fixed rankings and query records.

placement rule within that specific sidecar. However, PROTECTED minus NoFacet is -0.00743 $[-0.01616, 0.00132]$, statistically unresolved. In BGE-native space, PROTECTED minus Unprotected is $+0.003367$ $[-0.001285, 0.008026]$, and PROTECTED minus NoFacet is $+0.000631$ $[-0.004220, 0.005365]$; both are statistically unresolved. Component effects are therefore system-dependent.

5.3 Comparison with a Strong Dense Retriever

The pre-specified BGE baseline outperforms original FULL. The dataset-equal-weight FULL minus BGE difference is -0.03998 $[-0.05393, -0.02621]$. The complete interval lies below zero,

Contrast	Δ F1 [95% CI]	State
MiniLM facet	+0.01336 [0.00341, 0.02343]	Supported
MiniLM protection	+0.00354 [-0.00675, 0.01389]	Unresolved
Sidecar placement	+0.01122 [0.00129, 0.02104]	Supported
Sidecar facet	-0.00743 [-0.01616, 0.00132]	Unresolved
Native placement	+0.00337 [-0.00129, 0.00803]	Unresolved
Native facet	+0.00063 [-0.00422, 0.00536]	Unresolved
Flat-ball control	—	Not defined

Table 2: Complete component evidence. “Native” denotes the BGE-native confirmation setting; effects are not transported between semantic spaces. The flat-ball row has no matched estimand.

distinguishing this negative result from a statistically unresolved contrast. On this evaluation, the original MiniLM-based HyperGranular-RAG system performs below the BGE baseline.

On the same joint component setting, BM25

Setting	Δ F1 [95% CI]	State
Gemma transfer	+0.00516 [−0.00262, 0.01295]	Unresolved
Full vs. BM25	−0.00755 [−0.02259, 0.00773]	Unresolved
Full vs. hybrid	−0.00320 [−0.01591, 0.00942]	Unresolved
Full vs. BGE	−0.03998 [−0.05393, −0.02621]	Negative
Sidecar vs. BGE	−0.00256 [−0.00998, 0.00458]	Unresolved
Native vs. BGE	−0.00305 [−0.00688, 0.00063]	Unresolved

Table 3: Strong-retriever comparisons and generator transfer. Each row uses its own evaluation setting; rows are not pooled.

and a Dense–BM25 hybrid provide additional reference points. FULL minus BM25 is -0.00755 $[-0.02259, 0.00773]$, and FULL minus a Dense–BM25 hybrid is -0.00320 $[-0.01591, 0.00942]$. Both comparisons are statistically unresolved. BGE is the stronger reference point because the entire FULL-minus-BGE interval is negative.

5.4 Cross-Space Sidecar Evaluation

The cross-space sidecar asks whether the fixed MiniLM structural expansion can add complementary candidates while BGE retains the primary ranking. Its core PROTECTED minus BGE difference is -0.00256 $[-0.00998, 0.00458]$. The sidecar comparison therefore remains statistically unresolved relative to BGE. The positive protected-placement contrast answers a different question: better placement of a fixed inserted set does not show that the set itself improves BGE.

5.5 BGE-Native Reconstruction

The BGE-native study removes the cross-space mismatch by rebuilding candidate geometry and facets in BGE space. Its confirmation comparison remains statistically unresolved. The dataset-equal-weight PROTECTED minus BGE answer-F1 difference is -0.003046 $[-0.006880, 0.000631]$; the EM difference is -0.003667 $[-0.007667, 0.000004]$. Dataset-specific F1 differences are -0.002907 $[-0.008682, 0.002652]$ for HotpotQA and -0.003184 $[-0.008486, 0.001863]$ for MuSiQue. Both point estimates are slightly negative, but all intervals cross zero. The tested BGE-native reconstruction therefore provides no statistically resolved answer-quality improvement over BGE in this setting.

The positive C10 development difference is not pooled with this result: development selected the configuration, and the separate confirmation split evaluated it.

5.6 Generator Transfer

Under the single pre-specified Gemma mobile-QAT configuration, the dataset-equal-weight FULL minus DENSE difference is $+0.00516$ $[-0.00262, 0.01295]$, which is statistically unresolved. This transfer result applies only to the tested Qwen FP16 and Gemma mobile-QAT deployment configurations on the recorded RTX 4060 Laptop 8GB, fixed short-answer prompt, and 4,096-token cap.

5.7 Efficiency and Evidence Displacement

These measurements describe the recorded hardware rather than cross-device efficiency. For the BGE-native confirmation run, retrieval reconstruction took 8.544 seconds; BGE and BGE-native PROTECTED generation took 887.650 and 884.198 seconds, respectively; recorded GPU peak memory was 3.592 GiB.

Post hoc supporting-evidence analysis clarifies why coverage and answer quality diverge (Figure 4). In the BGE-native PROTECTED condition, HotpotQA adds 3 gold units, displaces 2, and has net $+1$; MuSiQue adds 16, displaces 15, and also has net $+1$. Nevertheless, answer-F1 point estimates relative to BGE are negative. In the cross-space sidecar, HotpotQA has 21 added, 29 displaced, and net -8 , whereas MuSiQue has 59 added, 50 displaced, and net $+9$. These counts are post-decision descriptive analyses and were not used to retune the method.

6 Discussion

Why compact retrieval can benefit. The repeated MiniLM gains support a focused interpretation: when a compact representation leaves recoverable evidence gaps, local candidate organization plus query-conditioned cross-ball expansion can expose complementary units that a flat ranking omits. The joint component confirmation’s facet contrast supports facet-mediated candidate selection within that implementation. We therefore interpret the repeated gains as evidence that structured completion can recover complementary units omitted by this compact retriever.

Why the increment contracts under strong BGE. BGE begins with substantially higher answer quality than the original MiniLM system. The original FULL ranking is clearly worse than BGE, and both tested extensions around BGE yield narrow intervals around small negative point

estimates. A stronger representation may already rank much of the recoverable evidence highly, leaving less marginal opportunity for structural expansion. Additional units then compete with already strong evidence. These results show that the marginal value of the tested structural completion contracts with BGE; they do not determine whether other structure-aware approaches can complement strong retrievers.

Evidence addition is not evidence utility. The BGE-native confirmation is especially informative: net gold evidence increases slightly, but answer quality does not. A gold-labelled unit can be redundant with retained context, weakly positioned, difficult for the generator to use, or offset by a different displacement. Conversely, a non-gold unit can affect the answer. CR/ER, gold transitions, and answer F1 therefore answer different questions.

Placement matters but cannot substitute for efficacy. The cross-space sidecar placement result isolates ranking position because protected and unprotected conditions contain the same inserted set. It supports preserving high-ranked evidence under a fixed budget. However, the BGE-native confirmation does not independently confirm the placement increment, and neither placement experiment establishes that PROTECTED exceeds BGE.

Facet effects depend on semantic space. Facet-conditioned selection beats the centroid-only No-Facet arm in the fixed MiniLM system, but its increments are statistically unresolved in cross-space and BGE-native studies. Query-to-ball geometry, seed identities, facet eligibility, and competition with the base ranking all depend on representation space. The MiniLM effect is therefore neither transported to BGE nor interpreted as superiority over untested diversity/coverage selectors.

Selection and displacement explain failure modes. Post-label controller analyses locate failure at selection and displacement: resource budgeting retained harm more often than gain, the query-level score reversed direction, and a candidate-level probe remained below its advancement gate (Table 6). These development results are explanatory, not confirmation; complementary-candidate detection remains open.

Separating development from confirmation. The positive BGE-native development result mo-

tivated one pre-specified confirmation evaluation. It did not license retroactive configuration search after confirmation. We therefore report the confirmation result without further tuning on the confirmation set.

Interpreting statistically unresolved comparisons. The sidecar, BGE-native, and generator-transfer intervals cross zero. Without a pre-specified equivalence margin and an equivalence test, these results cannot be called equal. They also do not establish harm.

7 Limitations

Evaluation is closed-candidate rather than full-wiki or open-domain; it does not measure indexing, approximate-nearest-neighbor recall, or corpus-scale noise. The repeated positive evidence uses one Qwen generator, and the only additional generator is a single Gemma mobile-QAT configuration. Only one strong BGE backbone is tested. The BGE-native configuration family is finite and development-selected; untested variants remain unknown.

A scientifically matched flat granular-ball control was not defined, so the granular-ball structure lacks an independent ablation claim. Protected insertion has inconsistent independent evidence: cross-space sidecar placement is supported, while the joint-component and BGE-native protection-related contrasts are statistically unresolved. Facet evidence is representation- and implementation-dependent. No matched relevance-diversity, maximum-coverage, or other simple completion selector was evaluated, so the hyperedge representation has not been shown necessary or superior to such alternatives.

The HotpotQA training split may have appeared in generator pretraining. The project can guarantee only that its own retrieval and evaluation respected the pre-specified data separation. Post hoc mechanism analyses are descriptive, not confirmatory causal analyses. Sample sizes reflect available resources and interval precision, not a formal power design. No statistically unresolved result is evidence of equivalence.

8 Reproducibility

The repository records exact dataset and model revisions, configurations, rankings, predictions, and evaluation outputs. Independent scripts reconstruct the reported metrics and decisions from

these artifacts. Every plot has a machine-readable source, and the manuscript build is deterministic. Licenses are recorded separately; caches and third-party weights are excluded.

9 Conclusion

HyperGranular-RAG provides repeatable answer-F1 gains for the compact MiniLM retriever across three closed-candidate evaluations, and facet-conditioned selection improves over its centroid-only comparator. The BGE experiments show that this increment is backbone-dependent: the original MiniLM-based system performs below BGE, while the cross-space and BGE-native extensions remain statistically unresolved. Protected placement can improve the ordering of a fixed candidate set, but candidate selection remains central to end-to-end answer quality. Overall, structured evidence completion is useful in the evaluated compact retrieval setting, while its value with stronger retrievers remains an open methodological problem.

Ethical Considerations

This study uses existing benchmark datasets and publicly documented model artifacts. It does not recruit human participants or collect new personal data. The analysis reports negative, statistically unresolved, and unmatched results alongside positive findings to reduce selective-reporting risk. A venue-specific determination of whether a formal institutional ethics statement is required remains subject to human confirmation.

Data and Code Availability

The identity-exposing development repository records code, configurations, frozen derived evaluation artifacts, figure source data, and verification manifests. It is not linked in this anonymous draft. Original benchmark datasets, model weights, and large local caches are not redistributed and must be obtained from their official sources under their respective terms. An anonymous artifact snapshot and final reuse license remain pending human decisions.

Generative AI Disclosure (Draft)

Generative AI tools were used for code assistance, experiment-workflow assistance, documentation assistance, and language-editing assistance.

They are not authors and do not bear academic responsibility. Human authors retain responsibility for the scientific design, evidence selection, verification, claims, and submitted text. Final wording will be reviewed against the selected venue’s policy before submission.

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A Deterministic Procedures and Fixed Parameters	800
Algorithm 1: adaptive granular-ball construction. Initialize a queue with $(U_q, 0)$. For each ball B at depth z , compute c_B and r_B . If the depth, size, and radius/size gates do not permit a split, emit B . Otherwise choose the first seed as the unit farthest from c_B , choose the second as the unit least similar to the first, and assign each unit to its more similar seed. Candidate-order ties go to the earlier seed. Accept the split only if both children have at least two units; otherwise emit B . Continue until the queue is empty. Stable candidate order and <code>unit_id</code> resolve remaining ties.	801
Algorithm 2: query-aware facet-hyperedge selection. Tokenize the query and every ball with the frozen rules, intersect each ball’s term set with the query-term set, and select the two highest query-to-centroid balls as seeds. Compute newly covered terms and the frozen h_B score for every non-seed ball. Reject balls that fail any eligibility gate. Sort survivors by $(-h_B, \text{edge_id})$, retain at most two distinct balls, and order their units by original MiniLM cosine and <code>unit_id</code> . No Gold label or answer score enters this procedure.	802
Algorithm 3: protected bounded insertion. Let $K_q = \min(20, U_q)$ and protect the first	803

$\min(10, K_q)$ unique Dense units. Traverse the eligible expansion order, retain units at or above the frozen score floor, skip protected duplicates, and place at most four units immediately after the prefix. Append unused Dense units in their original order, then expansion units only if needed to fill K_q . Deduplicate stably and truncate to K_q .

Complexity. For $n = |U_q|$, $B = |\mathcal{B}_q|$, embedding dimension d , and maximum depth H , Dense scoring/sorting costs $O(nd + n \log n)$; recursive two-seed assignment costs $O(Hnd)$; facet scoring and ordering costs $O(n + Bd + B \log B)$ apart from tokenization; insertion costs $O(n)$. Working memory is $O(nd + n + B)$. Encoder and generator inference are excluded.

B Scope of Applicability

C Absolute Strong-Retriever Outcomes

D Analysis of Failure Modes

E Efficiency and Verification Summary

F Scope of Supported Claims

Group	Fixed value	Provenance and sensitivity status
Ball split	depth < 6; splittable size $\geq$ 4; radius gate 0.78; minimum child size 2	Frozen compact implementation; no confirmatory sensitivity sweep
Ball seeds and ties	farthest from centroid; then least similar to seed 1; stable candidate order	Deterministic rule, not learned
Facet seeds/budget	2 seed balls; at most 2 expansion balls	Frozen compact implementation; no confirmatory budget sweep
Facet gates	new terms $\geq$ 1; ball size $\leq$ 8; $b_B \geq .12$ or $a_B \geq .05$; units/new term $\leq$ 6; redundancy $\leq .90$; $h_B \geq .10$	Frozen compact implementation
Facet score weights	.45, .20, .20, .15, -.20, -.05; unit bonus 0	Frozen implementation weights; no claim of global optimality
Insertion	protected prefix 10; budget 4; effective $K = 20$	Frozen transferred policy
Candidate floor	0.1957079917192459	Stage2E calibration 25th percentile, transferred exactly; not an independently established optimum
Ordering	descending cosine or h_B , then unit_id or edge_id	Fully deterministic tie-breaking

Table 4: Fixed compact-method parameters. “No confirmatory sensitivity sweep” is a limitation, not evidence of robustness. BGE-native development used a separate finite configuration family and is not pooled with this table.

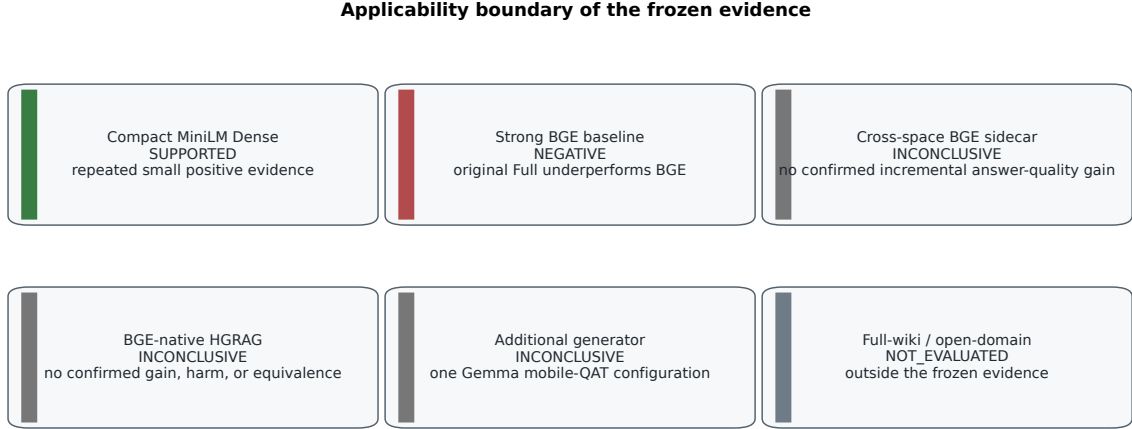


Figure 3: **Scope of applicability.** Repeated small gains are established only for the compact MiniLM backbone under closed-candidate Top-20 evaluation. The original system is negative against strong BGE, the tested sidecar and BGE-native extensions are statistically unresolved, and full-wiki/open-domain operation was not evaluated.

Evaluation setting	Method	F1	EM	CR@20	ER@20
Joint component (Stage4H)	BGE	.35801	.30467	.85400	.93902
Joint component (Stage4H)	Original FULL	.31803	.26400	.72217	.87657
Cross-space sidecar (Stage4I)	BGE	.34301	.28000	.84967	.93594
Cross-space sidecar (Stage4I)	PROTECTED	.34045	.27767	.84350	.93459
BGE-native confirmation (Stage5A)	BGE	.34569	.28500	.86167	.94288
BGE-native confirmation (Stage5A)	PROTECTED	.34264	.28133	.86233	.94288

Table 5: Dataset-equal-weight absolute outcomes in the strong-retriever evaluations. Comparisons are valid only within the same setting; rows from different settings are not pooled.

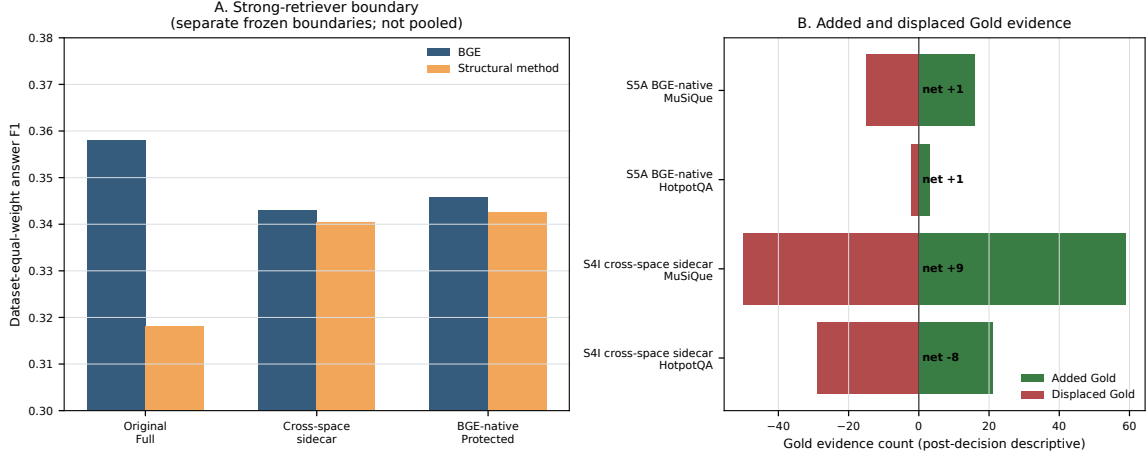


Figure 4: **Strong-retriever comparisons and evidence displacement.** Panel A compares each structural method only with BGE inside its own Stage4H, Stage4I, or Stage5A setting; the groups are not pooled. Panel B reports post-decision descriptive Gold evidence added and displaced. These counts did not control selection or advancement and do not replace end-to-end answer evaluation.

Layer	Observation	Evidential role
Static expansion	Repeated compact-MiniLM answer-F1 gains, but query-level gain and harm coexist	Confirmatory, bounded
Resource controller	Retained 47/94 gain and 53/69 harm queries; retention gap -0.26812 ; two of six gates passed	Valid negative development result
Query score	Gain-vs-harm AUROC 0.39269 [0.30558, 0.48150]	Exploratory, direction reversed
Candidate probe	AUROC 0.64310 [0.55520, 0.72974], AP 0.72198; Brier 0.26321 vs. 0.25338 prevalence baseline	Exploratory, unresolved
Strong-retriever displacement	Small or mixed net Gold changes did not yield positive answer-F1 relative to BGE	Post-decision descriptive

Table 6: Failure analysis from Stage4B–Stage4D and the strong-retriever evaluations. Exploratory and post-decision rows explain observed failure modes but do not create new confirmatory claims.

Item	Value	Evidential role
Stage5A retrieval reconstruction	8.544 s	Hardware-bound descriptive
Stage5A BGE generation	887.650 s	Hardware-bound descriptive
Stage5A native Protected generation	884.198 s	Hardware-bound descriptive
Stage5A GPU peak memory	3.592 GiB	Hardware-bound descriptive
Bootstrap	10,000 iterations per frozen decision	Confirmatory
Gold policy	Isolated until rankings/predictions were frozen	Integrity
Verification	Independent final verifiers passed	Integrity

Table 7: Efficiency and verification summary. Time and memory values do not support cross-device efficiency claims.

Claim	Evidence state	Setting
Compact FULL exceeds DENSE in three evaluation settings	Supported	MiniLM, Qwen, closed-candidate Top-20
Facet increment in the fixed MiniLM system	Supported	Stage4H only
Protected placement with identical sidecar insertions	Supported	Stage4I only
Original FULL exceeds strong BGE	Negative	Stage4H
Cross-space sidecar exceeds BGE	Unresolved	Stage4I
BGE-native PROTECTED exceeds BGE	Unresolved	Stage5A
BGE-native placement increment	Unresolved	Stage5A
BGE-native facet increment	Unresolved	Stage5A
Additional-generator transfer	Unresolved	One Gemma mobile-QAT setting
Flat granular-ball control	Unmatched	No matched estimand
Granular balls are necessary	Not established	Flat contrast undefined
Facet hyperedges beat generic diversity/coverage	Not tested	No matched alternative selector
Open-domain deployment value	Not tested	Closed candidates only

Table 8: Scope of the manuscript’s supported conclusions. Statistically unresolved rows are not equivalence claims, and system-specific component evidence is not transported across representation spaces.